

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO3: Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)

Assessment Tool Weightage: 50%

Time Duration: 1 Hour

QUESTIONS: 4

Total Marks:40

Annexure A

CASE STUDY

Code of Conduct:

I B Agneeshwaran (192524508) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: AGNEESHWARAN B

Annexure A

CASE STUDY

Case Study 1 — Smart Medical Diagnosis System

A hospital is designing a rule-based Logical Agent for preliminary patient diagnosis. The system's knowledge base contains:

- I. If a patient has Fever AND Cough $\rightarrow$ Possible Flu
- II. If a patient has Fever AND Rash $\rightarrow$ Possible Measles
- III. If a patient has Cough AND Breathlessness $\rightarrow$ Possible Pneumonia
- IV. Facts about Patient A: Fever = True, Cough = True, Breathlessness = True

(a) Represent the above rules and facts precisely using Propositional Logic symbols. Define each symbol used.

(b) Apply Forward Chaining on the knowledge base to derive all possible diagnoses for Patient A. Show each inference step with the rule fired and fact derived.

(c) The goal is to confirm 'Patient A has Pneumonia.' Apply Backward Chaining and draw the goal reduction tree to verify this goal.

Case Study 2 — Autonomous Traffic Management Agent

A smart city traffic controller uses a FOL-based knowledge base to manage emergency vehicle routing:

- I. $\forall x: \text{Vehicle}(x) \wedge \text{EmergencyType}(x) \rightarrow \text{ClearPath}(x)$ [Rule 1]
- II. $\forall x: \text{ClearPath}(x) \rightarrow \text{GreenSignal}(x) \wedge \neg \text{RedSignal}(x)$ [Rule 2]
- III. $\forall x \forall y: \text{Vehicle}(x) \wedge \text{Behind}(x,y) \wedge \text{GreenSignal}(y) \rightarrow \text{Proceed}(x)$ [Rule 3]
- IV. Facts: $\text{Vehicle}(\text{Ambulance}), \text{EmergencyType}(\text{Ambulance}), \text{Vehicle}(\text{CarA}), \text{Behind}(\text{CarA}, \text{Ambulance})$

(a) Apply Unification to match Rule 1 with the given facts. Show the complete substitution set θ (MGU) at each step.

(b) Using the Resolution method, prove that 'Proceed (CarA)' is true. Convert relevant clauses to CNF and show each resolution step.

(c) Analyze whether the current FOL knowledge base is sufficient to handle two simultaneous emergency vehicles. Identify what additional rules or facts are needed and justify logically.

Case Study 3 — Agricultural Expert Reasoning System

An agricultural advisory logical agent operates with the following propositional knowledge base (KB):

- 1) P1: $\text{SoilDry} \rightarrow \text{IrrigationNeeded}$
- 2) P2: $\text{IrrigationNeeded} \wedge \text{CropWheat} \rightarrow \text{ApplyDripMethod}$
- 3) P3: $\neg \text{ApplyDripMethod} \wedge \text{CropWheat} \rightarrow \text{CropAtRisk}$
- 4) Facts: $\text{SoilDry} = \text{True}, \text{CropWheat} = \text{True}$

(a) Convert all sentences P1, P2, P3 and the facts into Conjunctive Normal Form (CNF). Show each conversion step (eliminate implication $\rightarrow$ move $\neg$ inward $\rightarrow$ distribute).

(b) Using Resolution Refutation, prove the theorem 'ApplyDripMethod'. Negate the goal, combine with the KB in CNF, and derive the empty clause $\square$. Draw the resolution proof tree.

(c) If the fact 'SoilDry' is removed from the KB, analyze step-by-step how logical conclusions change. Does 'CropAtRisk' still follow? Apply appropriate reasoning patterns to justify.

Case Study 4 — Wumpus World Knowledge Agent

An AI agent navigating the Wumpus World perceives: Stench at [1,2], Breeze at [1,1], Glitter at [2,2].

The agent uses a propositional KB with these rules:

- 1) Stench [1,2] $\rightarrow$ Wumpus is adjacent to cell [1,2]
- 2) Breeze [1,1] $\rightarrow$ Pit is adjacent to cell [1,1]
- 3) Glitter [x,y] $\rightarrow$ Gold is at [x,y]
- 4) $\neg$ Wumpus [x,y] $\wedge$ $\neg$ Pit[x,y] $\rightarrow$ Safe[x,y]

(a) Construct the agent's belief state from its percepts. Apply Modus Ponens systematically to derive new knowledge facts. List all derived conclusions.

(b) Using Forward Chaining from the agent's percepts, identify the possible locations of the Wumpus and Pit. Show each reasoning step precisely.

(c) Evaluate whether the path [1,1] $\rightarrow$ [1,2] $\rightarrow$ [2,2] is safe for the agent to collect the gold. Use the knowledge derived in (a) and (b) to justify your conclusion with logical inference.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding ; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

ANSWERS

Case Study 1 — Smart Medical Diagnosis System

(a) Propositional Logic Representation

Symbols Definition

- F = Patient has Fever
- C = Patient has Cough
- R = Patient has Rash
- B = Patient has Breathlessness
- Flu = Patient has Flu
- Measles = Patient has Measles
- Pneumonia = Patient has Pneumonia

Rules in Propositional Logic

1. $(F \wedge C) \rightarrow \text{Flu}$
2. $(F \wedge R) \rightarrow \text{Measles}$
3. $(C \wedge B) \rightarrow \text{Pneumonia}$

Facts

- F = True
- C = True
- B = True

(b) Forward Chaining

Step-by-Step Inference

Step 1:

Facts: F, C, B

Step 2: Apply Rule 1

$(F \wedge C) \rightarrow \text{Flu}$

$F = \text{True}, C = \text{True} \Rightarrow \text{Flu} = \text{True}$

Derived Fact: Flu

Step 3: Apply Rule 3

$(C \wedge B) \rightarrow \text{Pneumonia}$

$C = \text{True}, B = \text{True} \Rightarrow \text{Pneumonia} = \text{True}$

Derived Fact: Pneumonia

Step 4: Apply Rule 2

$(F \wedge R) \rightarrow \text{Measles}$

R is not given $\Rightarrow$ Cannot infer Measles

Final Diagnoses

- Flu
- Pneumonia

(c) Backward Chaining (Goal: Pneumonia)

Goal Reduction Tree

Goal: Pneumonia

$\rightarrow$ Using Rule 3: $(C \wedge B) \rightarrow \text{Pneumonia}$

Subgoals:

- $C (\text{True})$
- $B (\text{True})$

Since both subgoals are satisfied from facts,

Goal Proven: Pneumonia = True

Case Study 2 — Autonomous Traffic Management Agent

(a) Unification (Rule 1)

Rule 1

$\forall x: \text{Vehicle}(x) \wedge \text{EmergencyType}(x) \rightarrow \text{ClearPath}(x)$

Facts

- Vehicle(Ambulance)
- EmergencyType(Ambulance)

Unification Process

Match:

- $x = \text{Ambulance}$

MGU (Most General Unifier)

$\theta = \{x / \text{Ambulance}\}$

Result

ClearPath(Ambulance)

(b) Resolution Proof for Proceed(CarA)

Convert to CNF

Rule 1:

$\neg \text{Vehicle}(x) \vee \neg \text{EmergencyType}(x) \vee \text{ClearPath}(x)$

Rule 2:

$\neg \text{ClearPath}(x) \vee \text{GreenSignal}(x)$

$\neg \text{ClearPath}(x) \vee \neg \text{RedSignal}(x)$

Rule 3:

$\neg \text{Vehicle}(x) \vee \neg \text{Behind}(x,y) \vee \neg \text{GreenSignal}(y) \vee \text{Proceed}(x)$

Facts in CNF

Vehicle(Ambulance)

EmergencyType(Ambulance)

Vehicle(CarA)

Behind(CarA, Ambulance)

Resolution Steps

1. From Rule 1 + Facts
→ ClearPath(Ambulance)
2. From Rule 2
→ GreenSignal(Ambulance)
3. Apply Rule 3

Vehicle(CarA), Behind(CarA, Ambulance), GreenSignal(Ambulance)

→ Proceed(CarA)

Conclusion

Proceed(CarA) is TRUE

(c) Analysis for Multiple Emergency Vehicles

Problem

Current KB handles only single emergency vehicle.

Missing Elements

- Priority rules for multiple emergency vehicles
- Conflict resolution rules
- Time-based scheduling

Required Additional Rules

1. $\text{Priority}(E1) > \text{Priority}(E2) \rightarrow \text{ClearPath}(E1)$ first
2. If two vehicles: assign alternate routing
3. Traffic segmentation rules

Conclusion

Current KB is **insufficient** for multiple emergencies.

Case Study 3 — Agricultural Expert System

(a) CNF Conversion

P1: SoilDry → IrrigationNeeded

$\neg \text{SoilDry} \vee \text{IrrigationNeeded}$

P2: IrrigationNeeded ∧ CropWheat → ApplyDripMethod

$\neg \text{IrrigationNeeded} \vee \neg \text{CropWheat} \vee \text{ApplyDripMethod}$

P3: $\neg \text{ApplyDripMethod} \wedge \text{CropWheat} \rightarrow \text{CropAtRisk}$

$\text{ApplyDripMethod} \vee \neg \text{CropWheat} \vee \text{CropAtRisk}$

Facts

SoilDry

CropWheat

(b) Resolution Refutation (Prove ApplyDripMethod)

Negate Goal

$\neg \text{ApplyDripMethod}$

Steps

1. SoilDry
2. $\neg \text{SoilDry} \vee \text{IrrigationNeeded}$
→ IrrigationNeeded
3. $\neg \text{IrrigationNeeded} \vee \neg \text{CropWheat} \vee \text{ApplyDripMethod}$
→ ApplyDripMethod
4. $\text{ApplyDripMethod} + \neg \text{ApplyDripMethod}$
→ $\square$ (Empty Clause)

Conclusion

ApplyDripMethod is TRUE

(c) Removing SoilDry

Effect

- Cannot derive IrrigationNeeded
- Cannot derive ApplyDripMethod

Check CropAtRisk

Needs:

$\neg \text{ApplyDripMethod} \wedge \text{CropWheat}$

Since ApplyDripMethod not derived → uncertain

Conclusion

CropAtRisk cannot be guaranteed

Case Study 4 — Wumpus World Agent

(a) Belief State and Modus Ponens

Given Percepts

- Stench(1,2)
- Breeze(1,1)
- Glitter(2,2)

Derived Knowledge

1. Stench(1,2) → Wumpus adjacent
→ Possible Wumpus: (1,1), (1,3), (2,2)
2. Breeze(1,1) → Pit adjacent
→ Possible Pit: (1,2), (2,1)
3. Glitter(2,2) → Gold(2,2)

(b) Forward Chaining

Wumpus Locations

From Stench:

- (1,1), (1,3), (2,2)

Pt Locations

From Breeze:

- (1,2), (2,1)

(c) Path Safety Analysis

Path

$[1,1] \rightarrow [1,2] \rightarrow [2,2]$

Evaluation

- (1,2): Possible Pit → Dangerous
- (2,2): Possible Wumpus → Dangerous

Conclusion

Path is **NOT SAFE**